

Lesson 05 Practice: CPU, ALU, Control Unit & Registers

JKSSB Computer Awareness

October 10, 2026

Provenance: 20 original JKSSB-pattern questions (*[Original]*) written for this lesson; none reproduces an official paper item.

Q1. *[Original]* Expand ALU.

- A. Arithmetic Logic Unit
- B. Advanced Logic Unit
- C. Arithmetic Language Unit
- D. Application Logic Unit

Q2. *[Original]* Which CPU component performs arithmetic and logical operations on data?

- A. Control Unit
- B. Register file
- C. Arithmetic Logic Unit (ALU)
- D. System bus

Q3. *[Original]* The main function of the Control Unit is to:

- A. perform addition and subtraction of numbers
- B. direct the sequence of operations and issue control signals
- C. store intermediate arithmetic results
- D. provide permanent secondary storage

Q4. *[Original]* Which of the following is **NOT** a function of the Control Unit?

- A. Decoding instructions
- B. Issuing control signals
- C. Directing the flow of data between units
- D. Performing the addition of two numbers

Q5. *[Original]* Which register holds the address of the **next** instruction to be fetched?

- A. Instruction Register (IR)
- B. Program Counter (PC)
- C. Memory Data Register (MDR)
- D. Accumulator (ACC)

- Q6.** *[Original]* Which register holds the instruction that is currently being decoded and executed?
- A. Program Counter (PC)
 - B. Memory Data Register (MDR)
 - C. Instruction Register (IR)
 - D. Memory Address Register (MAR)
- Q7.** *[Original]* Which register holds the address of the memory location to be accessed?
- A. Memory Data Register (MDR)
 - B. Memory Address Register (MAR)
 - C. Instruction Register (IR)
 - D. Accumulator (ACC)
- Q8.** *[Original]* Which register holds the data being transferred to or from memory?
- A. Memory Address Register (MAR)
 - B. Program Counter (PC)
 - C. Memory Data Register (MDR)
 - D. Instruction Register (IR)
- Q9.** *[Original]* The accumulator (ACC) is primarily used to hold:
- A. the address of the next instruction
 - B. intermediate arithmetic and logic results
 - C. the last instruction fetched from memory
 - D. a permanent copy of the operating system
- Q10.** *[Original]* Which is the fastest storage in a computer system?
- A. Registers
 - B. L1 cache
 - C. RAM
 - D. Solid-state drive
- Q11.** *[Original]* Where are registers located?
- A. In cache memory
 - B. In main memory (RAM)
 - C. In secondary storage
 - D. Inside the CPU
- Q12.** *[Original]* Clock speed is measured in hertz (Hz). A processor rated at 2.4 GHz completes approximately:
- A. 2.4 million cycles per second
 - B. 2.4 billion cycles per second

- C. 2.4 thousand cycles per second
- D. 2.4 trillion cycles per second

Q13. *[Original]* Which statement about clock cycles is correct?

- A. Every machine instruction completes in exactly one clock cycle.
- B. All instructions require the same number of clock cycles.
- C. Different instructions may take different numbers of clock cycles.
- D. A clock cycle is measured in bytes.

Q14. *[Original]* Evaluate the statements:

- I. The ALU works on binary data using logic gates.
- II. The Control Unit directs the sequence of operations and data flow.
- III. The Control Unit performs the arithmetic calculations.

- A. I only
- B. I and II only
- C. II and III only
- D. I, II, and III

Q15. *[Original]* Which abbreviation is correctly matched with its full form?

- A. PC — Program Counter
- B. IR — Instruction Recorder
- C. MAR — Memory Arithmetic Register
- D. MDR — Main Data Register

Q16. *[Original]* The Program Counter holds the address of the _____ instruction to be fetched.

- A. next
- B. current
- C. previously executed
- D. data

Q17. *[Original]* Registers used for general temporary storage of operands and intermediate results, without a fixed dedicated role, are called:

- A. status registers
- B. general-purpose registers
- C. memory address registers
- D. instruction registers

Q18. *[Original]* The status (flags) register holds:

- A. the address of the memory location to access

- B. the instruction to be decoded
 - C. condition flags such as zero, carry, and sign
 - D. the data to be written to memory
- Q19.** *[Original]* A quad-core processor means that the processor:
- A. has four bytes of cache memory
 - B. supports four monitors at once
 - C. runs four operating systems
 - D. contains four independent processing cores on a single chip
- Q20.** *[Original]* In the office desktop, which unit fetches each instruction, decodes it, and issues control signals to the ALU, registers, and memory?
- A. Arithmetic Logic Unit
 - B. Control Unit
 - C. Hard disk
 - D. Monitor
- Q21.** *[JKSSB FAA-pattern]* Which register holds the address of the next instruction to be executed?
- A. Program Counter
 - B. Stack Pointer
 - C. Data Register
 - D. Printer Buffer
- Q22.** *[JKPSI-pattern]* MAR stands for:
- A. Memory Address Register
 - B. Main Access Router
 - C. Machine Arithmetic Rule
 - D. Monitor Alignment Record
- Q23.** *[SSC CGL-pattern]* Which part of the CPU performs comparison operations such as greater-than checks?
- A. ALU
 - B. Monitor
 - C. Hard disk
 - D. Speaker
- Q24.** *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which of the following are CPU registers?
- A. Program Counter (PC)
 - B. Instruction Register (IR)
 - C. Memory Address Register (MAR)

D. USB port

Q25. [SSC CGL-pattern] (MSQ: select ALL correct options.) Which statements about CPU internals are correct?

- A. The Program Counter holds the next instruction's address.
- B. The Instruction Register holds the current instruction.
- C. The ALU carries out arithmetic and logical operations.
- D. The Control Unit stores user files permanently.

Q26. [PYQ-adapted: SSC CGL 2025 Tier-II Paper-I, 19-Jan-2026 Q6; SRC-18] Which register holds the instruction currently being decoded?

- A. Memory Address Register
- B. Program Counter
- C. Instruction Register
- D. Accumulator

Q27. [PYQ-adapted: JKSSB Junior Assistant 2026 Q64, final key; SRC-03/SRC-06] Which register normally holds the address of the next instruction to be fetched?

- A. Instruction Register
- B. Program Counter
- C. Accumulator
- D. Memory Data Register

Q28. [Original] Which register temporarily holds the address of a memory location being accessed?

- A. MAR
- B. IR
- C. Accumulator
- D. Status register

Q29. [Original] The clock rate of a processor is commonly expressed in:

- A. Bytes per second
- B. Hertz
- C. Pixels per inch
- D. Revolutions per minute

Q30. [Original, MSQ drill] (MSQ: select ALL correct options.) Match each register with its common role.

- A. PC — address of the next instruction
- B. IR — current instruction being decoded
- C. MAR — memory address for a transfer
- D. Accumulator — intermediate arithmetic/logic result